

Call for Expressions of Interest	
<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	IBM Bob GeoQuest AI: Generate a 3D Adventure Game from a Real-World Location
<i>Company:</i>	IBM
Contact Details	
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Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☐ If Applicable: Please provide details
<i>Relevant research areas (if applicable)¹:</i>	Game AI; procedural content generation; geospatial computing; data engineering; human-computer interaction; software engineering; simulation; pathfinding; open data; AI-assisted development; digital skills development.
Project Background Information	
This project asks students to build a playable 3D adventure game that can generate an explorable world from a selected real-world map area, such as Doncaster town centre or another chosen location. The goal is not to copy Google Maps or any proprietary imagery. Instead, students should use open geospatial data and procedural generation to create a recognisable, game-ready approximation of roads, buildings, paths, parks, rivers and mission zones. The project combines software engineering, open data, procedural world generation and game AI. Students will design a pipeline that imports map data, converts it into a simplified 3D scene, adds gameplay rules, and enables a player to explore the generated environment. This may include quests, collectible objects, puzzles, checkpoints, AI-controlled non-player characters, enemies, route-based missions or educational challenges based on the local geography. IBM Bob should be used as the development accelerator throughout the project lifecycle. Students should use IBM Bob to understand geospatial data structures, design the software architecture, generate and debug code, create procedural generation algorithms, improve NPC behaviours, document the project, test edge cases and explain their engineering	

¹ An overview of different research areas in the school of computer science can be found [here](#).

decisions. The project should remain zero-cost by using open-source software, open data and original or openly licensed assets only.

A successful outcome would be a visual, demonstrable software project suitable for a university showcase, IBM student engagement activity, hackathon or portfolio demonstration. It should help students show practical AI-assisted software engineering skills while creating an engaging digital experience that is easy for non-technical audiences to understand.

Requirements

- A playable 3D adventure game prototype built in a zero-cost game/web framework such as Godot, Unity Personal, Three.js, Babylon.js or another agreed open-source/academic-friendly toolset.
- A location selection mechanism allowing users to choose a bounded map area, for example by entering a place name, coordinates, bounding box or selecting a prepared sample location.
- A data ingestion pipeline that uses open geospatial data to obtain roads, paths, buildings, water, parks and other relevant map features. The project must not scrape Google Maps imagery or require paid map APIs for the core deliverable.
- A procedural world-generation module that converts map features into game-ready 3D geometry, including simplified roads, buildings, navigation paths, spawn points and mission zones.
- A basic player exploration mode with movement, camera controls, collision handling and interaction with objects or characters.
- At least three gameplay mechanics, such as quests, collectibles, puzzles, enemy/NPC behaviour, checkpoint navigation, timed challenges, route-finding missions or location-based story events.
- A clear AI/game logic component, such as pathfinding, NPC decision-making, procedural quest generation, adaptive difficulty, route planning or AI-assisted mission design.
- Documented IBM Bob usage showing how the team used IBM Bob for code generation, debugging, refactoring, architecture decisions, testing, documentation and gameplay design. This should include sample prompts and reflections on what was improved.
- A technical report explaining the architecture, data sources, data pipeline, game engine choice, AI/game logic, testing method, limitations, ethical considerations and future enhancements.
- A final demonstration video showing a selected location being transformed into a playable 3D adventure environment and explored in-game.
- A GitHub repository or equivalent code package containing source code, setup instructions, sample data, attribution for open assets, and a repeatable build/run process.

Skills

Highly desirable:

- **Strong programming fundamentals in Python, JavaScript, C# or GDScript.**
- **Experience with a game engine or 3D web framework.**
- **Understanding of data structures, APIs, file parsing and software architecture.**
- **Interest in AI-assisted software development using IBM Bob.**
- **Ability to work with Git/GitHub and produce clear technical documentation.**
- **Problem-solving mindset for debugging, testing and iterative prototyping.**

Desirable:

- Familiarity with OpenStreetMap or other open geospatial datasets.
- Procedural generation, pathfinding, navigation meshes or game AI.
- 3D modelling concepts, coordinate systems, vectors or basic geometry.
- UI/UX design and accessible gameplay design.
- Experience with open-source assets, attribution and licensing discipline.
- Interest in IBM Granite, watsonx or IBM SkillsBuild learning pathways.

Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.